NEURO FORM 1 by Karim Samir

1.c

Drug-induced hypersensitivity syndrome : Anticonvulsant hypersensitivity syndrome (AHS) is a rare but serious event associated with aromatic antiepileptics, including phenytoin. AHS represents an idiosyncratic drug reaction that occurs in 1 in 1000 to 1 in 10,000 patients treated with phenytoin and has a fatality rate of 10 percent. Symptoms vary but classically include fever, rash, lymphadenopathy, and possibly other organ involvement. Pharyngitis is common. When severe, the syndrome can include hepatitis, megaloblastic anemia, rhabdomyolysis, and arteritis

Signs of AHS are usually seen within two months of starting phenytoin therapy.

- **Lymph node biopsy** — Biopsy is appropriate if an abnormal node has not resolved after four weeks and should be performed promptly in patients with other findings suggesting malignancy (eg, rapid increase in size of the node; systemic complaints of fever, night sweats, weight loss).

the cause is obvious .. phenytoin must be stopped

2.d

subarachnoid hage

3.g

typical case of meniere

hear loss (sensorineural type) , hissing sound (tinnitus)
vomiting …. Etc

4.e

endolymphatic fistula
Barotrauma .. airplane

barotrauma due to increased pressure in airplane===== fistula ==== vertigo

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5.b

CT findings along ventricular margin (periventricular calcification)

6.e Uncal herniation

intracranial hage ==== edema and herniation

uncal herniation syndrome. herniation of the temporal uncus.

Oculomotor nerve compression ===== **causing the pupil of the affected eye to dilate and fail to constrict in response to light as it should**

ipsilateral third cranial nerve palsy (pupillary dilation, downward and outward eye deviation)

Hemiplegia due to compression of the cortical spinal tract in the midbrain often follows immediately

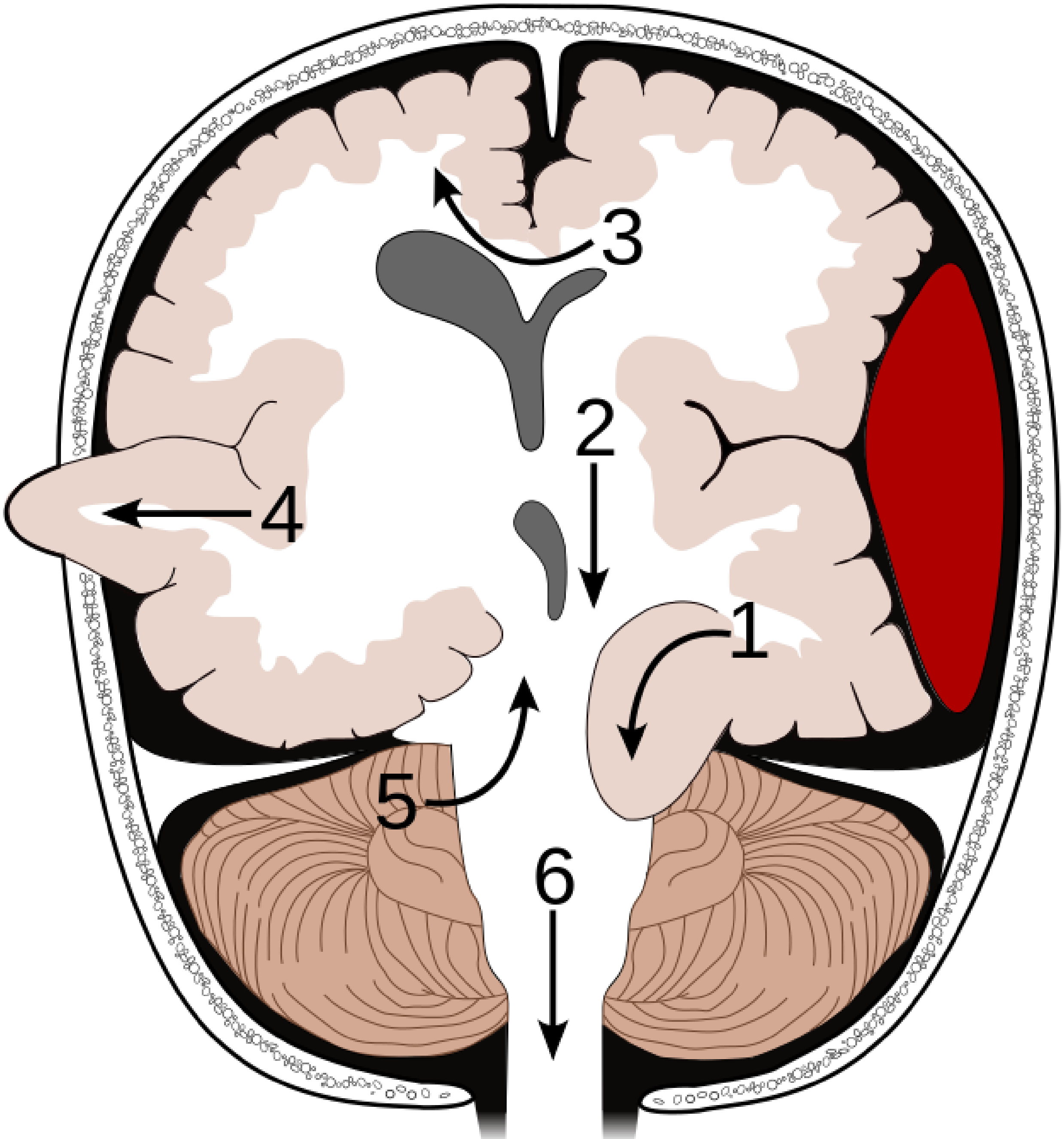
N.B

Supratentorial herniation

1. Uncal (transtentorial)
2. Central
3. Cingulate (subfalcine)
4. Transcalvarial
5. Tectal (posterior)

- **Infratentorial herniation**

5. Upward (upward cerebellar or upward transtentorial)
6. Tonsillar (downward cerebellar)



if u think of midbrain hage
the woman come with right arm and face numbness alarm signs (.. Middle cerebral artery territory ..
mid brain blood supply from vertebrobasilar system

7.c

Headaches, Urinary incontinence, Memory impairment, Seizures all these represent higher motor functions

8.b
trigeminal neuralgia

9.a
typical case of jakob disease

Myoclonic jerks and rapid progression of dementia

10.c

UTI …

Septic encephalopathy — Septic encephalopathy is the most common cause of acute toxic-metabolic encephalopathy (TME), and its presence and severity correlate with increased mortality [16,19-21]. The pathophysiology of septic encephalopathy is multifactorial. Microcirculatory abnormalities, altered blood-brain barrier permeability, inflammatory cytokines, reductions in monoamine neurotransmitters, and an increase in the concentration of the false neurotransmitter octopamine may all play a role

The cardinal feature of confusion and delirium is impaired attention; clinical findings can range from subtle cognitive difficulties to florid delirium or coma

Other common findings include **a disturbed sleep-wake cycle**, decreased alertness, hypervigilance, hallucinations, sensory misperceptions, impaired memory, and disorientation [1,5-7]. The thought process is often disorganized, manifested by confused or rambling conversation [5]. Paranoid ideation and excessive suspiciousness may occur. Affect is also compromised in TME; most patients tend to be apathetic and withdrawn, some seem anxious, agitated, and fearful, and others appear to be manic

management:
The treatment of TME focuses primarily on correcting the underlying condition

However, regardless of the cause of acute toxic-metabolic encephalopathy (TME), a number of general measures should be instituted :

-Review of medication list and discontinuation of all drugs with potential toxicity to the central nervous system, if possible
-Physical restraints should be used only as a last resort, if at all, as they frequently increase agitation and create additional problems, such as loss of mobility, pressure ulcers, aspiration, and prolonged delirium. In one study, restraint-use among patients in a medical inpatient unit was associated with a three-fold increased odds of persistent delirium at time of hospital discharge [3]. Alternatives to restraint use, such as constant observation (preferably by someone familiar to the patient such as a family member), may be more effective.
-Haloperidol may be given parenterally to treat severe agitation; **elderly subjects usually require only small doses**, such as 0.5 mg twice per day. Intravenous haloperidol has been associated with clinically significant QT prolongation requiring additional precautions regarding its use. Short-term use is advised

-Haloperidol should be avoided in cases of alcohol withdrawal, anticholinergic toxicity, and benzodiazepine withdrawal, and also in patients with parkinsonism.

-Thiamine should be administered to patients with a history of alcoholism, malnutrition, cancer, hyperemesis gravidarum, or renal failure on hemodialysis

11.e

mega doses of vitamins .. vit A

the headache and other symptoms … pseudotumor cerebri
Vitamin a toxicity

acute :
nausea, vomiting, vertigo, and blurry vision [39]. In very high doses, drowsiness, malaise, and recurrent vomiting can follow the initial symptoms

chronic :
Signs of chronic toxicity include ataxia, alopecia, hyperlipidemia, hepatotoxicity, bone and muscle pain, visual impairments, and many other non-specific signs and symptoms.

frontal lobe tumor : (increase ICP , propably beahavoiral also , pressure on olfactory and optic)

-Foster–Kennedy syndrome

12.b

lateral medullary syndrome .. posterior inferior cerebellar artery

Lateral medullary syndrome -
1) Vestibulocerebellar nuclei involvement - leading to unsteadiness and dizziness, vertigo and nytagmus
2) Nucleus Ambiguus(CN 9,10 nuclei) - difficulty swallowing
3) Descending spinothalamic tract - Ptosis and miosis of ipsilateral eye
4) Spinothalamic tract - decreased sensations on contralateral side of the body
5) Spinal nucleus of trigeminal - ipsilateral loss of sensations on face

N.B
when u see loss pain and temp ON one side of face and the opposite side of body
think of post inf cerebellar art on the side of face (it is classic finding for this artery)

13.b

typical case of bipolar…. Manic attack

14.e

neck stiffness .. 82 y .. osteoarthritis
pain from neck to left hand .. cervical radiculopathy
absent tricebs reflex .. c7 radiculopathy
the abnormal bone compress the c7 root make atrophy UL

abnormal bone compress on gracilis tract of posterior coloumn
the lower limb absent deep tendon reflexes

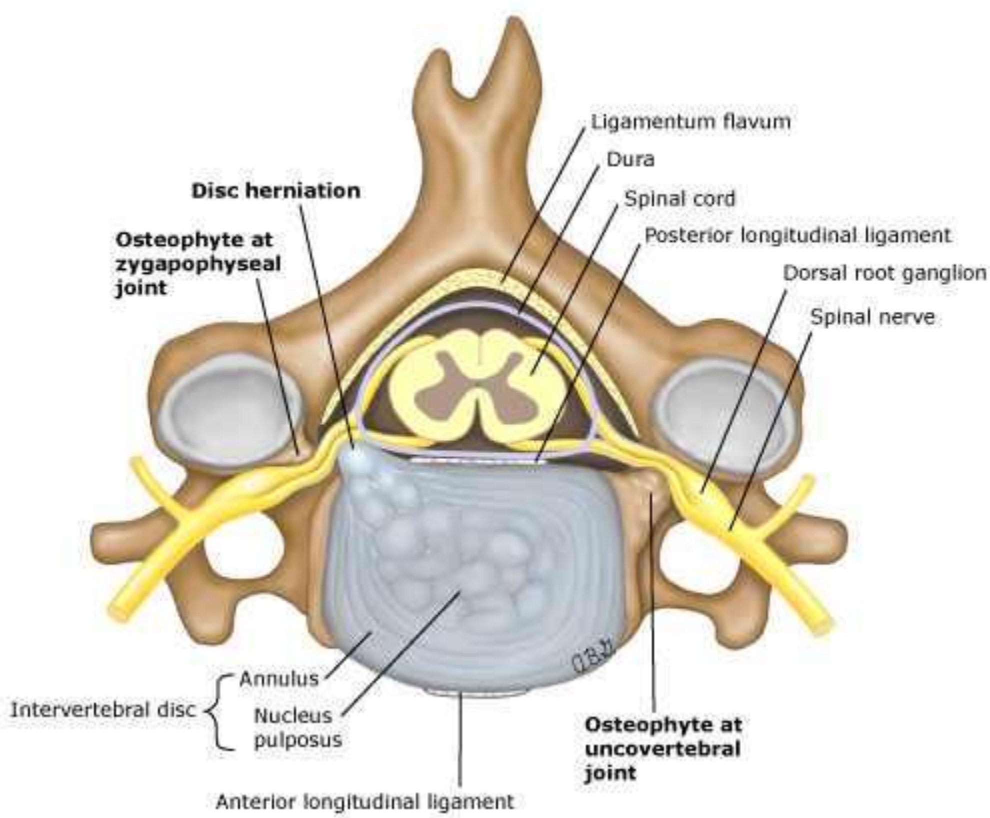
Cervical spondylotic myelopathy
Cervical spinal cord compression can give rise to weakness in the arms, legs, or both; sensory loss with a high thoracic or cervical sensory level; and urinary and rectal sphincter dysfunction.

There is no well-defined pattern of neurologic deficits in cervical spondylotic myelopathy. Patients present with varying signs and symptoms that can include:

- ●Pain in the neck, subscapular region, or shoulder, often radiating into the arms.
- ●Numbness or paresthesias in the arms. The sensory loss may follow a dermatomal pattern (figure 1), and often underlie complaints of loss of fine motor control in the hands.
- ●Gait disturbance, usually characterized by a spastic, scissoring quality.
- ●Sensory deficits, usually related to dorsal column function (reduced joint position and vibratory sense) and loss of pain sensation, can be elicited in the lower extremities and may contribute to the gait impairment.
- ●Weakness in the lower extremities with upper motor neuron characteristics: increased reflexes, increased tone, and present Babinski signs.
- ●Lower motor neuron findings (weakness, atrophy, and suppressed reflexes) in a myotomal distribution in the arms or hands. The C5-7 myotomes are most often affected (table 1).

●Bladder dysfunction, urgency, frequency, and/or retention.

●Lhermitte's sign. An electric shock-like sensation in the neck, radiating down the spine or into the arms, produced by forward flexion of the neck.



15.e

most of brain trauma or lesion can lead to … SIADH

right parietal ring enhanced lesion

16.a

ant.shoulder dislocation .. axillary N

17.c

recurrent ear infection .. some developmental delay in speaking

otits media with effusion
examination: retracted TM

Children with OME who have, or are at risk for, speech, language, or learning disorders should undergo hearing evaluation at the time of diagnosis of OME

then 2 choices :
watchful waiting and in most cases will resolve the effusion
or
tympanostomy : suggested at 6-12 months of continued bilateral OME or 4 months with bilateral hearing loss

18.a

corticospinal tract lesion … upper motor neuron …

19.b

compartment syndrome

20.b

methanol
home made liquor .. pallor of optic disc

organophosph poisoning .. DUMBLESS (diarrhea,urination, lacrimation … etc)

21.a

strock with no antiplatelet therapy === aspirin
strock and already on aspirin === aspirin +dypiridamol or clopidogril

U.WORLD Q ID 3712

22.c

23.a

Cauda equina syndrome -
1) Lower back pain + Unilateral radicular leg pain
2) Asymmetric sensory and motor findings in lower extremities
3) decreased/Absent lower extremity reflexes.
4) Saddle anesthesia, Urinary incontinence, decreased perianal sensations.

S1 nerve root radiculopathy - should cause weakness of foot plantar flexion, foot inversion, toe flexion, knee flexion and some hip extension. decreased sensations on posterior thigh and legs and decreased Ankle jerk. Not associated with decreased perianal sensations and urinary incontinence.

24.d (not 100 % sure b is an option too)

Brain imaging, preferably with magnetic resonance imaging (MRI), is indicated in the evaluation of patients with suspected AD
Brain MRI can document potential alternative or additional diagnoses including cerebrovascular disease, other structural diseases (chronic subdural hematoma, cerebral neoplasm, normal pressure hydrocephalus), and regional brain atrophy suggesting frontotemporal dementia or other types of neurodegenerative disease.

Structural MRI findings in AD include both generalized and focal atrophy, as well as white matter lesions. In general, these findings are nonspecific. The most characteristic focal finding in AD is reduced hippocampal volume or medial temporal lobe atrophy

25.a

rh.fever… mitral stenosis… irregular pulse…. AF …………… embolic accident to brain

26.a

alcoholic and phenytoin both decrease folic acid in few months
vit b12 need years to decrease

27.d

sore throat resolved without ttt … rheumatic Sydenham chorea

the onset of chorea usually occurs one to eight months after the inciting infection
chorea typically begins with distal movements of the hands, but generalized jerking of the face and feet may occur. The movements are rapid, irregular, and nonstereotypic jerks that are continuous while the patient is awake but improve with sleep.
Motor symptoms, including ballismus, facial grimacing, and gross fasciculations of the

tongue, often are observed in addition to frank chorea. These symptoms are associated with loss of fine motor control, muscle weakness, and hypotonia
Differential diagnosis : Tourette syndrome (Tics are sudden, brief, intermittent movements (motor tics) or utterances (vocal or phonic tics). Tics have been considered involuntary, but tics **can temporarily be voluntarily suppressed**)

TREATMENT — Sydenham chorea typically improves gradually, with a mean duration of 12 to 15 weeks
initial treatment for SC: prednisone
Treatment with either valproic acid or carbamazepine is a reasonable alternative

28.d

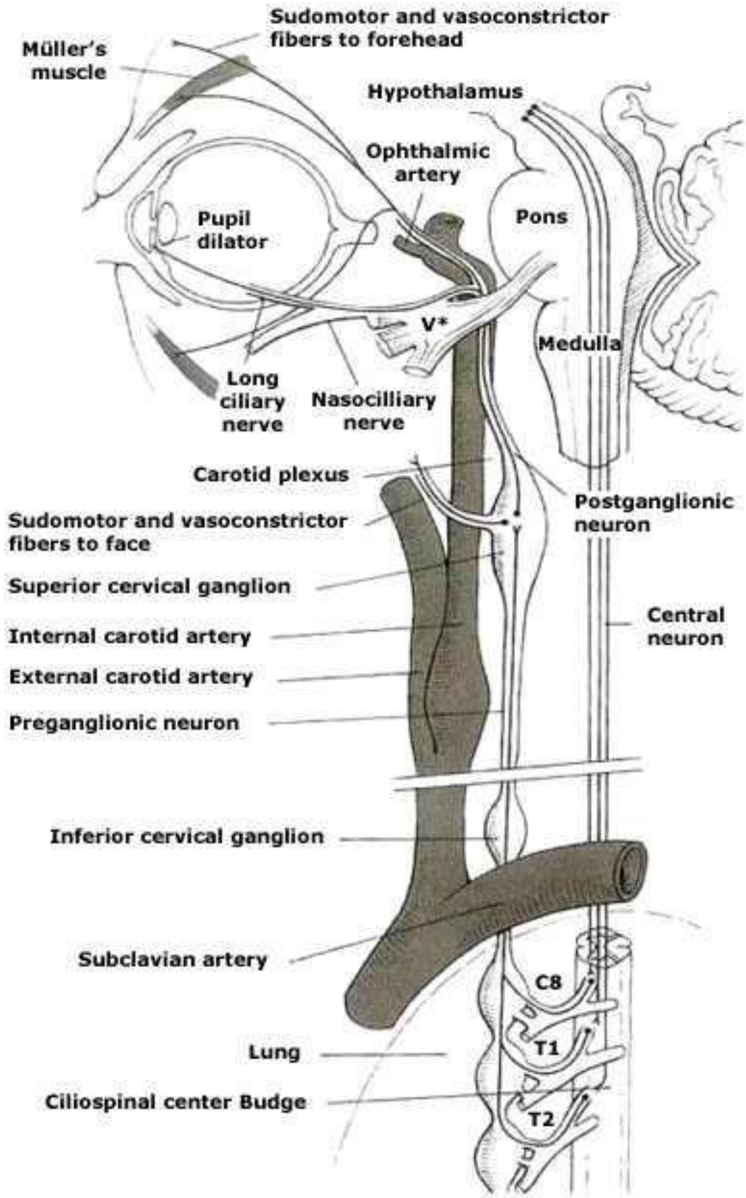
amaurosis ***fugax***
look for carotid stenosis by duplex as source of problem

29.c

Hospice care provides medical care and support to a patient with life-limiting illness and his/her family with a **focus on quality of life** rather than life prolongation or cure. The hospice philosophy embraces the general principle of **a comfortable death with dignity**. The care and treatment provided are **based on the patient's and family's goals and values.**
Hospice is appropriate (1) when patients are entering the last weeks to months of life AND (2) patients and their families decide to **forego disease-modifying or therapies with curative intent** (**the husband refused gastrostomy**) in order to focus on maximizing comfort and quality of life.

30.a

Horner syndrome
lower neck
other areas … we should notice more signs of other cranial nerves
unilateral Ptosis, Anhidrosis and miosis - Horners syndrome.



The first-order neuron descends caudally from the hypothalamus to the first synapse, which is located in the cervical spinal cord (levels C8-T2, also called ciliospinal center of Budge).

●The second-order neuron travels from the sympathetic trunk, through the brachial plexus, over the lung apex. It then ascends to the superior cervical ganglion, located near the angle of the mandible and the bifurcation of the common carotid artery.

●The third-order neuron then ascends within the adventitia of the internal carotid artery, through the cavernous sinus, where it is in close relation to the sixth cranial nerve [1]. The oculosympathetic pathway then joins the ophthalmic (V1) division of the fifth cranial nerve (trigeminal nerve). In the orbit and the eye, the oculosympathetic fibers innervate the iris dilator muscle as well as Müller's muscle, a small smooth muscle in the eyelids responsible for a minor portion of the upper lid elevation and lower lid retraction

First-order syndrome — Lesions of the sympathetic tracts in the brainstem .. The most common cause is a lateral medullary infarction

Second-order syndrome — Second-order or preganglionic Horner syndromes can occur with trauma or surgery involving the **spinal cord, thoracic outlet, or lung apex**

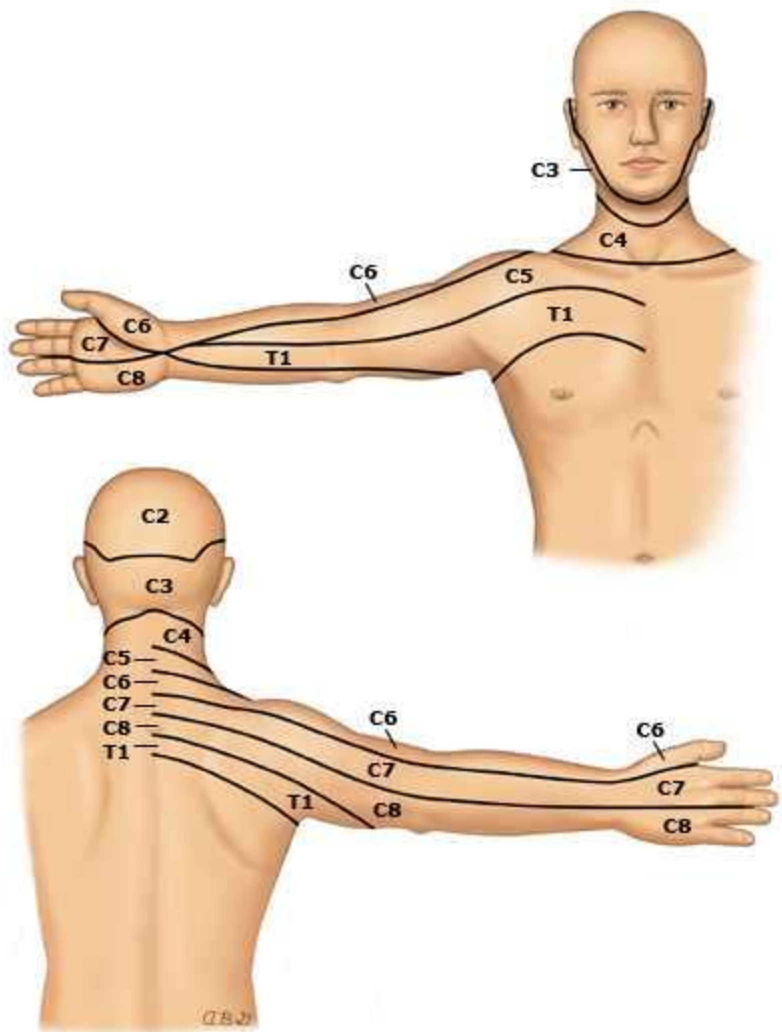
Third-order syndrome — Third-order Horner syndromes often indicate lesions of the internal carotid artery such as an arterial dissection, thrombosis, or cavernous sinus aneurysm

An acute Horner syndrome with neck or facial pain should be presumed to be caused by **carotid dissection** until proven otherwise

31.d

ulnar nerve compression

c8.. pain (**Neck, shoulder**, medial forearm, fourth and fifth digits, medial hand)



32.c

neurocystercosis

33.e

unpasteurized cheese
and monocyte 40%

34.d

UNCAL HERNIATION … transtentorial

compress cr.nerve 3 dilated pupil not reactive to light
already explained in previous question number 6

35.c

optic neuritis … multiple sclerosis .. young woman

36.d

37.a

Acute cerebellar ataxia is a clinical syndrome characterized by the sudden onset of ataxia, usually manifested as gait disturbance. Associated symptoms may include nystagmus, slurred or garbled speech, vomiting, irritability, dysarthria, or headache. Fever, meningismus, and seizures are absent. Most cases occur in toddlers or school-aged children, in many cases the symptoms develop a few days or weeks after a viral illness, particularly varicella.
CSF analysis in acute cerebellar ataxia is typically normal or shows a mild lymphocytic pleocytosis, with or without elevations in protein.
The prognosis for children with typical features of acute cerebellar ataxia is good. About 90 percent of patients have complete resolution of symptoms within a few weeks without specific treatment, while the remaining 10 percent have some long-term neurologic sequelae.

38.d

rheumatoid arthritis + corticosteroid for 10 years + old age
symmetrical weakness of dermatomal pattern both U.L
need investigate cervical spines with MRI

39.e

myethenia gravis … The initial therapy for most patients with myasthenia gravis (MG) is an oral anticholinesterase (ie, acetylcholinesterase inhibitor) medication, usually pyridostigmine

40.d

Wernicke encephalopathy
Encephalopathy : disorientation, indifference, and inattentiveness
Oculomotor dysfunction — Nystagmus, lateral rectus palsy, and conjugate gaze palsies reflect lesions of the oculomotor, abducens, and vestibular nuclei. Ocular abnormalities usually occur in combination rather than alone.
Gait ataxia — Ataxia primarily involves stance and gait and is likely due to a combination of polyneuropathy, cerebellar involvement, and vestibular dysfunction [1,54]. When severe, walking is impossible. Less affected **patients walk with a wide-based gait and slow**, short-spaced steps. Gait abnormalities are appreciated only on tandem gait in some patients.

41.d

tongue involvement suggest medullary lesion in the brain stem … tumor of the foramen magnum will compress

42.e

common peroneal n …. Foot drop, paresthesias and/or sensory loss over dorsum of foot and lateral shin
Weakness on foot dorsiflexion and eversion; sensory loss on dorsum of foot; reflexes normal

43.g

This pt has headache: aggravated by stress, noise; relieved by sleep and ACT. No aura, no GI symptoms, no neurological symptoms, and No other clues to suggest other Dx

44.e

polymyalgia + blindness … temporal arteritis

45.h

hypertensive emergency … nitroprusside

46.b

bruise behind the ear … temporal bone .. middle meningeal artery … epidural hematoma

47.d

essential tremors … ttt. Propranolol

48.d

major depression .. SSRI

49.h

sensation lost below umbilicus (t10)
probably epidural abscess at T8
u lose pain and temp below the lesion by 2 vertebra

50.a

prevent future strock if u have carotid stenosis above 50 by endarterectomy
below 50 antiplatelet therapy

ASPIRIN
she is allergic to it so .. clopidogrel

GENERAL :

- Since antiplatelet agents are effective for the prevention of recurrent ischemic stroke in patients with a history of noncardioembolic ischemic stroke or TIA of atherothrombotic, lacunar (small vessel occlusive), or cryptogenic type, nearly all such patients should be treated with an antiplatelet agent. Aspirin (50 to 100 mg daily), clopidogrel (75 mg daily), and the combination of aspirin-extended-release dipyridamole (25 mg/200 mg twice a day) are all acceptable options for preventing recurrent noncardioembolic ischemic stroke

- Long-term anticoagulation with warfarin, dabigatran, apixaban, or rivaroxaban should be considered as prevention for patients with chronic nonvalvular atrial fibrillation who have had an ischemic stroke or transient ischemic attack

- Antithrombotic therapy is the main treatment for patients with ischemic neurologic symptoms caused by **cervical artery dissection**. The choice between aspirin and anticoagulation in this setting remains somewhat controversial.